🔑 **Answer Key — Teacher Copy · Lesson 5-6**

MSTAR PRACTICE WORKSHEET

# Lesson 5-6: Represent Three-Dimensional Figures in Two Dimensions

Grade 6 Mathematics · Standard 6.GR.4

*These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.*

## Part A — Skills Check

One point each.

**Question 1** SELECTED RESPONSE — CORRECT: C

A rectangular prism has l = 6 in, w = 4 in, h = 2 in. What is its surface area?

A. 44 in²

B. 48 in²

C. 88 in² ✓

D. 88 in³

SA = 2(6×4) + 2(6×2) + 2(4×2) = 48 + 24 + 16 = 88 in².

**Question 2** SELECTED RESPONSE — CORRECT: B

How many faces does a rectangular prism have?

A. 4

B. 6 ✓

C. 8

D. 12

A rectangular prism has 6 faces: top, bottom, front, back, left, and right.

**Question 3** SELECTED RESPONSE — CORRECT: B

Surface area is measured in which type of unit?

A Cubic units (in³, cm³, ft³)

B Square units (in², cm², ft²)

C Linear units (in, cm, ft)

D No units needed

Surface area measures flat surfaces, so it uses square units. Volume uses cubic units.

**Question 4** SELECTED RESPONSE — CORRECT: C

A cube has edges of 5 cm. What is its surface area?

A 30 cm²

B 125 cm²

C 150 cm²

D 150 cm³

A cube has 6 identical faces. Each face: 5 × 5 = 25 cm². Total SA = 6 × 25 = 150 cm².

**Question 5** SELECTED RESPONSE — CORRECT: D

You unfold a time capsule that is a cube. Its net shows 6 equal squares, and each square has an area of 4 cm². What is the cube's total surface area?

A 10 cm²

B 12 cm²

C 16 cm²

D 24 cm²

A cube net has 6 equal square faces. With each face 4 cm², the surface area is 6 × 4 = 24 cm².

**Question 6** SELECTED RESPONSE — CORRECT: C

Your team unfolds a rectangular time capsule box into a flat net. How many faces (flat sides) will the net show in all?

A 4

B 5

C 6

D 8

A rectangular prism has 6 faces — top, bottom, front, back, left, and right — so its net shows 6 faces.

## Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

**Question 7** TWO-PART QUESTION

**Part A — correct answer: B**

A net folds into a rectangular prism. How many rectangular faces does that net have?

A 4

B 6

C 8

D 12

A rectangular prism has 6 faces — top and bottom, front and back, and two ends — so its net shows all 6.

- **A:** That counts only the four side faces, leaving out the top and the bottom.
- **C:** A rectangular prism has 8 vertices, but the net shows its flat faces. Count those.
- **D:** A rectangular prism has 12 edges, but the net shows its flat faces. Count those.

**Part B — correct answer: C**

Which statement must be true for a flat pattern to be a valid net of a solid?

A It must be a single long strip.

B Every face must be the same size.

C It folds into the solid with no faces overlapping and none missing.

D It must contain only triangles.

A net is a flat pattern that folds into the solid with every face present exactly once and no overlap.

**Question 8** SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about nets of three-dimensional figures:

- **A** A net shows every face of the solid laid flat.
- **B** A triangular prism net has 2 triangles and 3 rectangles.
- **C** The area of a net equals the surface area of the solid.
- **D** A net may have faces that overlap when folded.
- **E** One solid can have more than one correct net.

A, B, C, and E are true. D is false — overlapping faces mean the pattern does not fold into the solid, so it is not a net.

## Part C — Show Your Thinking

Model answer and scoring guidance.

**Question 9** WRITTEN RESPONSE · 2 POINTS

Two boxes have the same volume of 60 in³. Box A is 5 × 4 × 3. Box B is 10 × 6 × 1. Which box uses less wrapping paper (has less surface area)? Show your calculations and explain why a more cube-like shape might use less material.

**Model answer:** Both boxes hold 60 in³, but wrapping paper is measured by surface area. Box A: 2(20) + 2(15) + 2(12) = 94 in². Box B: 2(60) + 2(10) + 2(6) = 152 in². Box A uses less paper because its dimensions are closer to a cube — flattening the same volume into 10 × 6 × 1 creates two huge 60 in² faces and a lot of extra covering.

*Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.*

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Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.